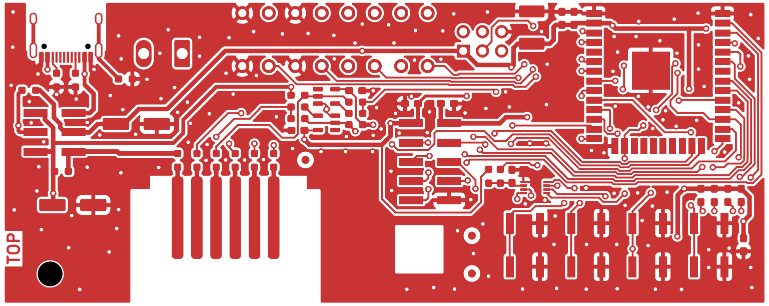
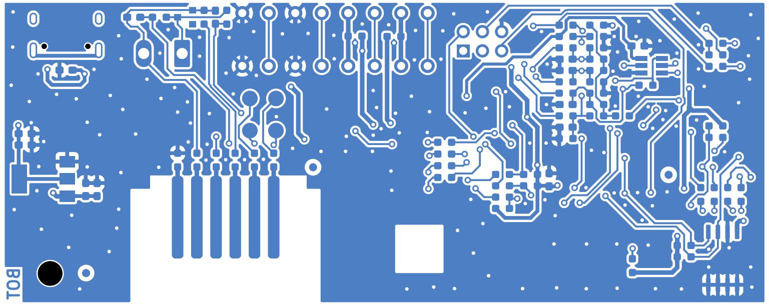
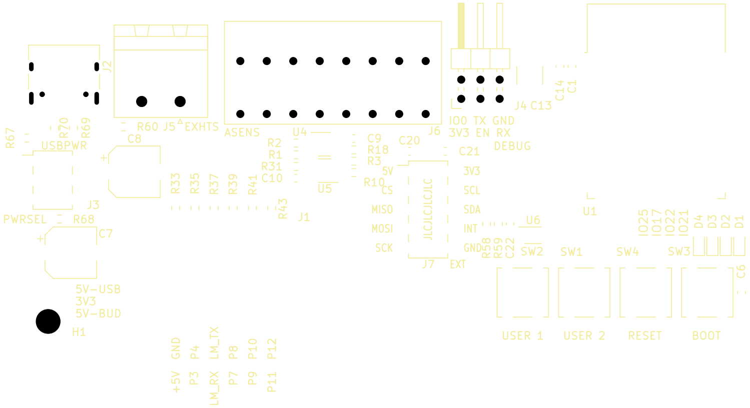
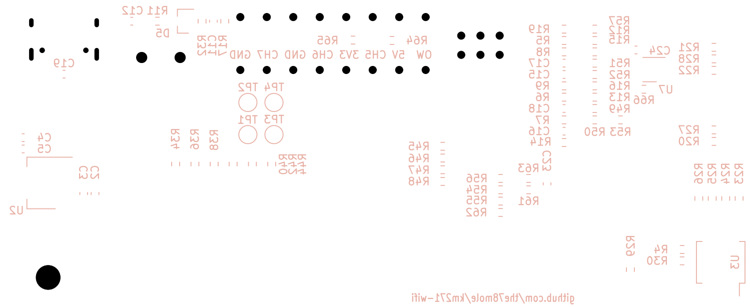
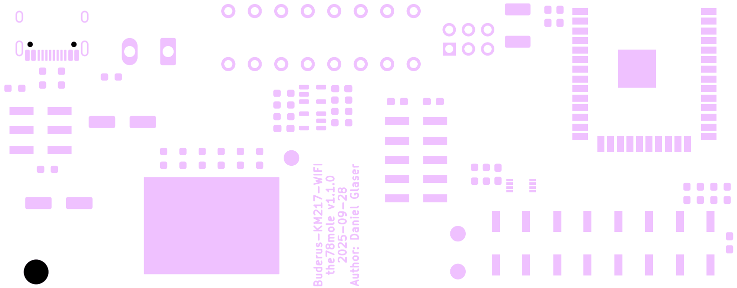

TOP

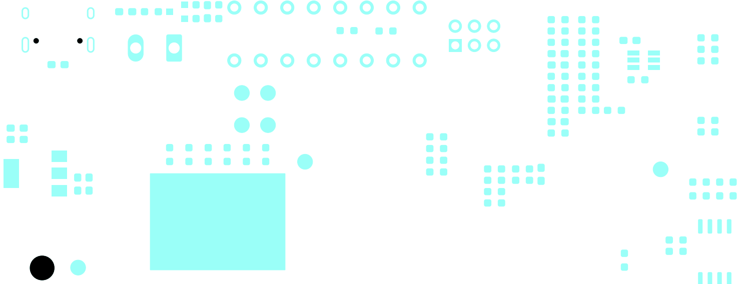

BOT

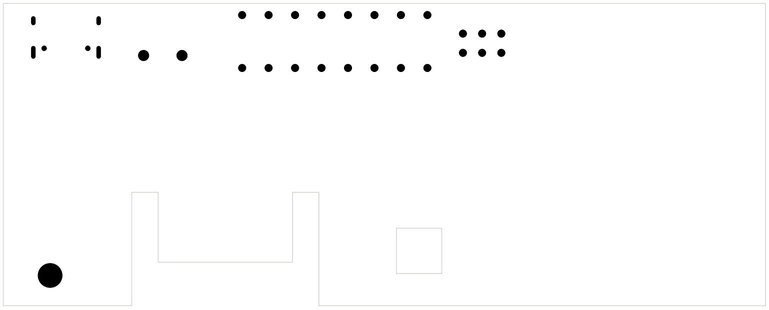

J2
R60
J5
EXHTS
ASENS
U4
J6
IO0
TX
GND
3V3
EN
RX
DEBUG
J4
C13
C14
C1
R67
R70
R69
USBPWR
C8
R2
R1
R31
C10
C9
R18
R3
R10
C20
C21
U5
5V
CS
MISO
MOSI
SCK
3V3
SCL
SDA
INT
GND
JLCJLCJLCJLC
J7
EXT
R33
R35
R37
R39
R41
R43
J1
J3
PWRSEL
R68
C7
5V-USB
3V3
5V-BUD
H1
GND
P4
LM_TX
P8
P10
P12
+5V
P3
LM_RX
P7
P9
P11
R58
R59
C22
U6
U1
SW2
SW1
SW4
SW3
IO25
IO17
IO22
IO21
D4
D3
D2
D1
C6
USER 1
USER 2
RESET
BOOT

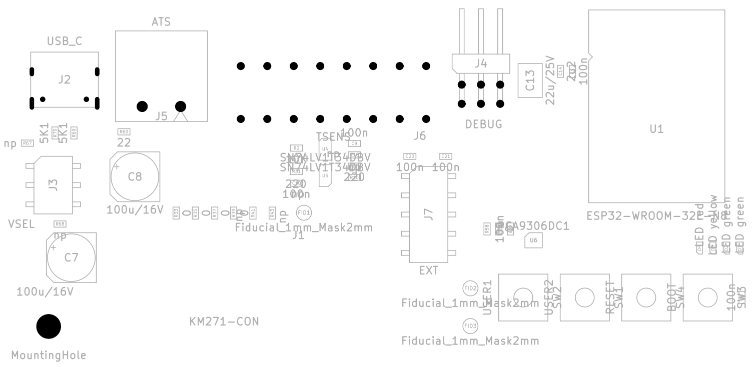


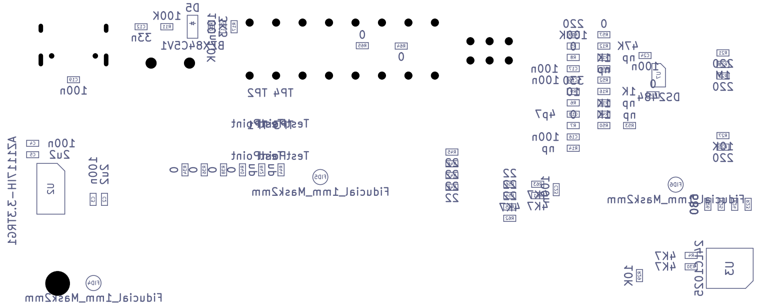
github.com/the78mole/km271-wifi
OW 5V CH5 3V3 CH6 GND CH7 GND
TP1
TP2
TP3
TP4
U2
U3
U7
D5

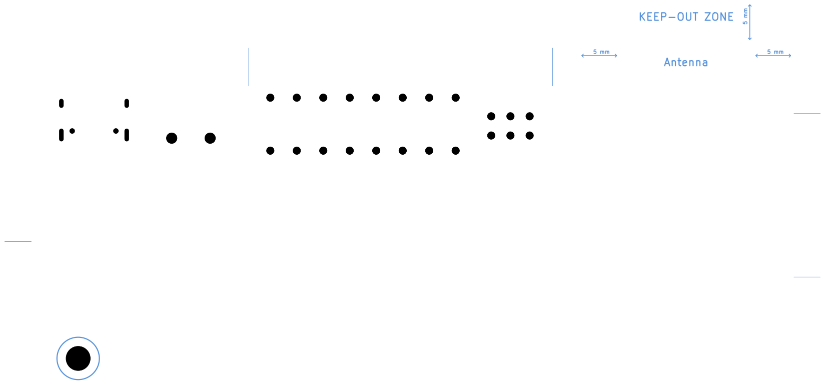

Buderus-KM217-WIFI
the78mole v1.1.0
2025-09-28
Author: Daniel Glaser






USB_C
J2
ATS
J5
J4
DEBUG
J6
C13
22u/25V
2u2
100n
U1
ESP32-WROOM-32E-N8
5K1
5K1
np
22
J3
VSEL
np
C8
100u/16V
C7
100u/16V
TSENS
100n
SN74LV1T34DBV
SN74LV1T34DBV
220
220
100n
np
0
0
0
np
np
Fiducial_1mm_Mask2mm
J1
100n
100n
J7
EXT
PCA9306DC1
U6
100n
LED red
LED yellow
LED green
LED green
USER1
USER2
SW2
RESET
SW1
BOOT
SW4
100n
SW3
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
KM271-CON
MountingHole





FAB NOTE J1 (KM271 card edge): 12 gold fingers F+B, 1.6 mm FR4 (+-10%).
Order option: Gold Fingers = YES (ENIG / hard gold on fingers), mask open over fingers.
Bevel: NOT POSSIBLE - finger edge (y=124.25) is recessed 5.75 mm between the board legs
(y=130) and JLC chamfers only boards >= 50x50 mm. Original module is used unbevelled.
Fingers end 0.2 mm before the board edge; no vias/pads within 1 mm (rule area GOLD_FINGERS).

FAB NOTES Buderus-KM217-WiFi rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +-10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: ENIG (gold fingers). Gold fingers: YES, bevel: NO (see J1 note)
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv